Geometry, Triangles & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Parallel Lines & Transversal Angle Relationships

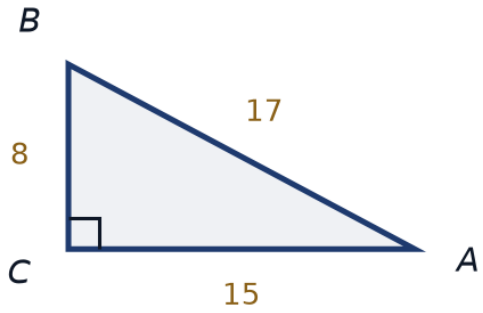
Two parallel lines $\ell_1 \parallel \ell_2$ are intersected by a transversal t. If a pair of consecutive interior (co-interior) angles have measures $(3x + 15)^\circ$ and $(2x - 5)^\circ$, find x and the measure of both angles.

2 [Foundational] • [FRQ] • Target: Triangle Angle-Sum & Exterior Angle Theorem

In ΔABC, the exterior angle at vertex C measures 134°, and the interior angle $\angle A$ measures 58°. Find $\angle B$ and $\angle ACB$.

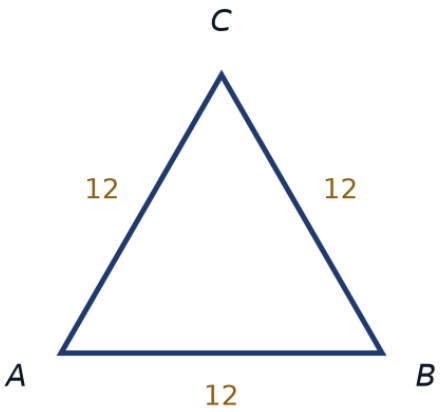
3 [Foundational] • [FRQ] • Target: Pythagorean Theorem & Right-Triangle Trig Ratios

In right triangle ABC shown below, $\angle C = 90^\circ$, $AC = 15$, and $BC = 8$. Find (a) the hypotenuse AB, (b) $\sin A$, (c) $\cos A$, and (d) $\tan B$.



4 [Foundational] • [FRQ] • Target: Special Right Triangles ($30^{\circ}\mathrm{-}60^{\circ}\mathrm{-}90^{\circ}$)

Find the exact perimeter and area of an equilateral triangle whose altitude has length $6\sqrt{3}$ cm.



5 [Foundational] • [FRQ] • Target: Special Right Triangles ($45^{\circ}\mathrm{-}45^{\circ}\mathrm{-}90^{\circ}$)

A square has diagonal of length $10\sqrt{2}$ m. Find its side length, perimeter, and area.

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6 [Foundational] • [FRQ] • Target: Triangle Inequality Theorem

Two sides of a triangle have lengths 7 and 12. State the strict inequality for all possible lengths x of the third side, and find how many integer values x can take.

7 [Foundational] • [FRQ] • Target: Isosceles Triangle Properties

In ΔPQR, $PQ = PR$ (isosceles) and the vertex angle $\angle P$ measures 44°. Find the measure of each base angle $\angle Q$ and $\angle R$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Similar Triangles & Parallel Cross-Sections

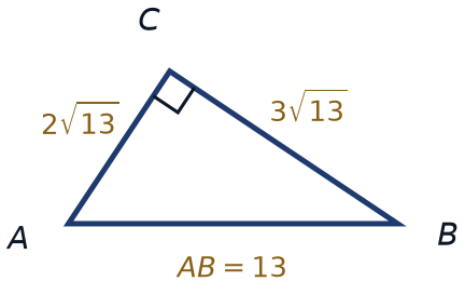
In ΔABC, segment DE is parallel to base BC with D on AB and E on AC. Given $AD = 6$, $DB = 9$, and $DE = 8$, find the length of BC and the ratio of the area of ΔADE to the area of ΔABC.

9 [Intermediate] • [FRQ] • Target: Complementary Angle Identity $\sin\theta = \cos(90^{\circ}-\theta)$

In right triangle DEF with $\angle F = 90^\circ$, $\sin D = \frac{5}{13}$. Find $\cos E$, $\cos D$, and $\tan E$.

10 [Intermediate] • [FRQ] • Target: Geometric Mean Theorems in Right Triangles

In right triangle ABC, the altitude CD is drawn to the hypotenuse AB. If $AD = 4$ and $DB = 9$, find the length of the altitude CD and both legs AC and BC.



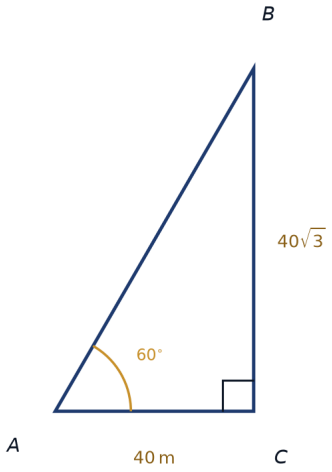
11 [Intermediate] • [FRQ] • Target: Interior & Exterior Angles of Regular Polygons

Find the measure of one interior angle and one exterior angle of a regular 15-sided polygon (pentadecagon).

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12 [Intermediate] • [FRQ] • Target: Angles of Elevation & Depression

A surveyor stands $40\,\text{m}$ from the base of a vertical radio tower on level ground. The angle of elevation to the top of the tower is 60°, and the surveyor's eye level is $1.6\,\text{m}$ above the ground. Find the exact height of the tower and its decimal approximation to two decimal places.

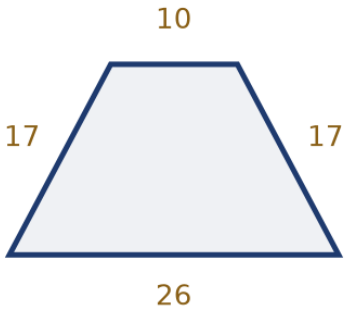


13 [Intermediate] • [FRQ] • Target: Triangle Angle Bisector Theorem

In ΔABC, the angle bisector of $\angle A$ meets side BC at point D. If $AB = 10$, $AC = 15$, and $BC = 20$, find the exact lengths of BD and DC.

14 [Intermediate] • [FRQ] • Target: Isosceles Trapezoids & Pythagorean Decomposition

An isosceles trapezoid has parallel bases of lengths $10\,\text{cm}$ and $26\,\text{cm}$ and non-parallel legs of length $17\,\text{cm}$. Find the height h, the area, and the length of each diagonal of the trapezoid.



Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Coordinate Triangle Geometry & Centroid • Cross-Lesson: Topic 1.01 (Coordinate Geometry)

In ΔABC, the vertices are $A(1, 2)$, $B(9, 2)$, and $C(5, 8)$. Verify that ΔABC is isosceles, find the length of the altitude from C to AB, and determine the coordinates of the centroid G.

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16 [Challenging] • [FRQ] • Target: Algebraic Right Triangles via Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Equations)

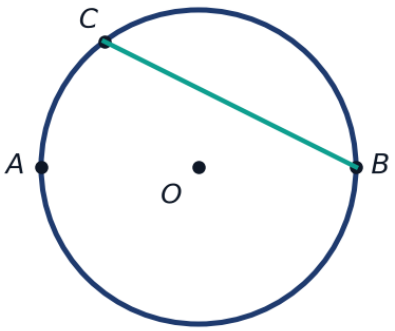
In a right triangle, one leg is $7\,\text{cm}$ longer than the other leg, and the hypotenuse is $17\,\text{cm}$. Set up and solve a quadratic equation to find the lengths of the two legs and the sine of the smaller acute angle.

17 [Challenging] • [FRQ] • Target: Law of Cosines & Trigonometric Triangle Area • Cross-Lesson: Topic 9.01 (Trigonometry)

In ΔABC, $AB = 7$, $BC = 8$, and $\angle B = 60^\circ$. Use the Law of Cosines to find the exact length of side AC and the exact area of ΔABC.

18 [Challenging] • [FRQ] • Target: Thales' Theorem & Inscribed Right Triangles • Cross-Lesson: Topic 9.01 (Circles & Inscribed Angles)

Triangle ABC is inscribed in a circle of diameter $AB = 20\,\text{cm}$. If $\cos A = \frac{3}{5}$, find the lengths of chords AC and BC and the area of the region inside the circle but outside ΔABC.



19 [Challenging] • [FRQ] • Target: Inscribed Square via Similar Right Triangles • Cross-Lesson: Topic 5.01 (Proportions) & Topic 8.01 (Area)

A square is inscribed inside a right triangle with legs 6 and 12 so that one vertex of the square coincides with the right-angle vertex of the triangle and the opposite vertex lies on the hypotenuse. Find the side length s of the square.

20 [Challenging] • [FRQ] • Target: Orthogonal Medians & Centroid Area Partition • Cross-Lesson: Topic 8.01 (Area Decomposition)

In ΔABC, medians AD and BE are perpendicular to each other and have lengths $AD = 9$ and $BE = 12$. Using the centroid ratio $2:1$, find the length of side AB and the area of ΔABC.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

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21 [Foundational] • [MCQ] • Target: Triangle Angle-Sum Theorem

In ΔABC, $\angle A = 42^\circ$ and $\angle B = 73^\circ$. What is the measure of $\angle C$?

a. 55°

b. 65°

c. 75°

d. 115°

22 [Foundational] • [MCQ] • Target: Pythagorean Triples

In a right triangle, the legs have lengths 9 and 12. What is the length of the hypotenuse?

a. 13

b. 15

c. 18

d. 21

23 [Foundational] • [MCQ] • Target: Complementary Trig Identities

If $\sin(3x - 10)^\circ = \cos(2x + 25)^\circ$, where both angles are acute, what is the value of x?

a. 13

b. 15

c. 17

d. 21

24 [Foundational] • [MCQ] • Target: Polygon Interior Angle Sum

What is the sum of the interior angles of a convex octagon (8 sides)?

a. 900°

b. 1080°

c. 1260°

d. 1440°

25 [Foundational] • [MCQ] • Target: Proportions in Similar Triangles

Triangles ABC and DEF are similar, with AB corresponding to DE. If $AB = 6$, $BC = 10$, and $DE = 9$, what is the length of EF?

a. 12

b. 13

c. 15

d. 18

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26 [Foundational] • [MCQ] • Target: Special Right Triangle Ratios

In a $30^\circ-60^\circ-90^\circ$ right triangle, the side opposite the 30° angle has length 7. What is the length of the side opposite the 60° angle?

a. $7\sqrt{2}$

b. $7\sqrt{3}$

c. 14

d. $\frac{7\sqrt{3}}{2}$

27 [Foundational] • [MCQ] • Target: Alternate Interior & Supplementary Angles

In the figure below, lines ℓ and m are parallel. If $\angle 1 = 128^\circ$, what is the measure of its alternate interior angle's supplement?

a. 38°

b. 52°

c. 62°

d. 128°

28 [Foundational] • [MCQ] • Target: Testing the Triangle Inequality

Which of the following sets of side lengths CANNOT form a triangle?

a. $4, 5, 8$

b. $6, 6, 11$

c. $5, 9, 14$

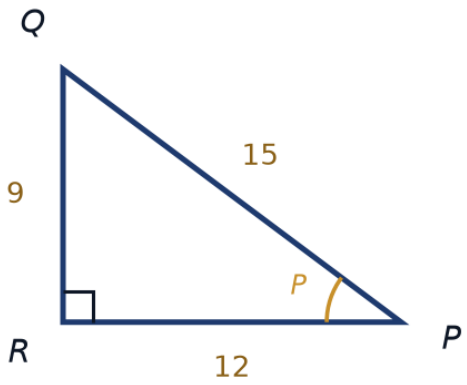
d. $7, 10, 12$

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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29 **[Intermediate]** • **[MCQ]** • **Target: Perimeter from Trigonometric Ratio**

In right triangle PQR shown below, $\angle R = 90^\circ$, $PR = 12$, and $\tan P = \frac{3}{4}$. What is the perimeter of ΔPQR?



a. 28

b. 32

c. 36

d. 48

30 **[Intermediate]** • **[MCQ]** • **Target: Area-to-Perimeter Ratio in Similar Triangles**

Two similar triangles have perimeters in the ratio $3 : 7$. If the area of the smaller triangle is $27\,\text{cm}^2$, what is the area of the larger triangle?

a. $63\,\text{cm}^2$

b. $108\,\text{cm}^2$

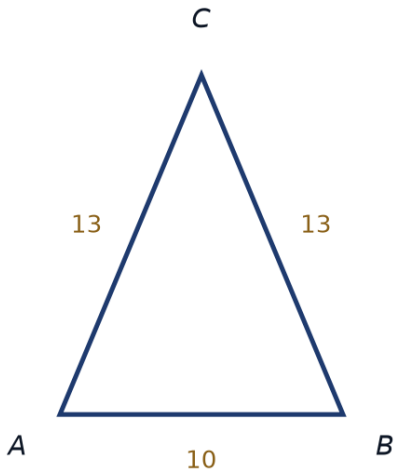
c. $147\,\text{cm}^2$

d. $189\,\text{cm}^2$

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31 [Intermediate] • [MCQ] • Target: Altitude & Area of Isosceles Triangles

In an isosceles triangle with two sides of length 13 and base of length 10, what is the area of the triangle?



a. 60

b. 65

c. 120

d. 130

32 [Intermediate] • [MCQ] • Target: Finding Number of Sides from Interior Angle

If each interior angle of a regular polygon measures 165°, how many sides does the polygon have?

a. 18

b. 20

c. 24

d. 30

33 [Intermediate] • [MCQ] • Target: Geometric Mean Theorem for Right-Triangle Legs

In right triangle ABC with $\angle C = 90^\circ$, altitude CD is drawn to hypotenuse AB. If $AC = 10$ and $AD = 5$, what is the length of AB?

a. 15

b. 20

c. 25

d. $5\sqrt{5}$

34 [Intermediate] • [MCQ] • Target: Sliding Ladder Pythagorean Word Problems

A 25-foot ladder leans against a vertical wall with its base 7 feet from the wall. If the top of the ladder slides down the wall by 4 feet, how many feet does the base of the ladder slide away from the wall?

a. 4

b. 7

c. 8

d. 15

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35 [Intermediate] • [MCQ] • Target: Midsegment Theorem & Area Partition

In ΔABC, D is the midpoint of AB and E is the midpoint of AC. If the area of trapezoid $DBCE$ is 45, what is the area of ΔABC?

a. 54

b. 60

c. 75

d. 90

36 [Intermediate] • [MCQ] • Target: Pythagorean Trigonometric Identities

If θ is an acute angle such that $\sin\theta = 3\cos\theta$, what is the value of $\sin\theta\cos\theta$?

a. $\frac{1}{10}$

b. $\frac{3}{10}$

c. $\frac{1}{3}$

d. $\frac{3}{5}$

37 [Intermediate] • [MCQ] • Target: Altitude Decomposition of Oblique Triangles

In ΔABC, $\angle A = 30^\circ$, $\angle C = 45^\circ$, and $AB = 12$. What is the exact length of BC?

a. $6\sqrt{2}$

b. $6\sqrt{3}$

c. $6\sqrt{6}$

d. $12\sqrt{2}$

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Inradius of a Right Triangle • Cross-Lesson: Topic 9.01 (Circles & Tangent Segments)

What is the radius of the circle inscribed in a right triangle with legs 9 and 40?

a. 3.5

b. 4

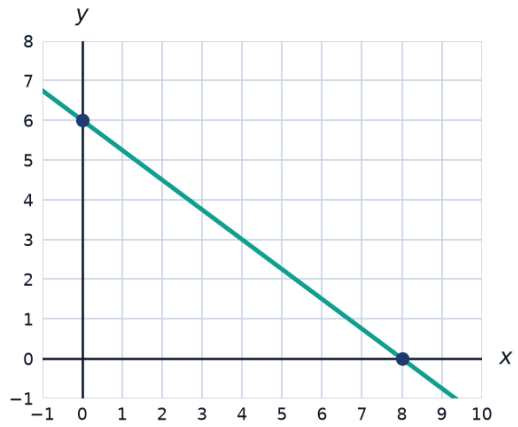
c. 4.5

d. 5

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39 [Challenging] • [MCQ] • Target: Slope as Tangent & Trigonometric Ratios of Lines • Cross-Lesson: Topic 1.01 (Linear Intercepts & Slope)

In the xy-plane, line $y = -\frac{3}{4}x + 6$ forms a right triangle with the positive coordinate axes. What is the value of $\sin\theta$, where θ is the acute angle the line makes with the x-axis?



a. $\frac{3}{5}$

b. $\frac{4}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{3}$

40 [Challenging] • [MCQ] • Target: Heron's Formula & 13-14-15 Triangle Altitude • Cross-Lesson: Topic 8.01 (Area of Triangles)

In ΔABC, sides a, b, c have lengths $13, 14, 15$. What is the length of the altitude to the side of length 14?

a. 10

b. 11

c. 12

d. 12.5

41 [Challenging] • [MCQ] • Target: Algebraic Squaring with $\sin^2\theta + \cos^2\theta = 1$ • Cross-Lesson: Topic 3.01 (Quadratic Identities)

If $\sin\theta + \cos\theta = \frac{7}{5}$ for an acute angle θ, what is the value of $\sin\theta\cos\theta$?

a. $\frac{12}{25}$

b. $\frac{24}{25}$

c. $\frac{7}{25}$

d. $\frac{12}{5}$

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42 [Challenging] • [MCQ] • Target: Isoperimetric Area Ratio of Regular Polygons • Cross-Lesson: Topic 5.01 (Ratios) & Topic 8.01 (Polygon Area)

A regular hexagon and an equilateral triangle have the exact same perimeter. What is the ratio of the area of the regular hexagon to the area of the equilateral triangle?

a. $1:1$

b. $3:2$

c. $2:1$

d. $4:3$

43 [Challenging] • [MCQ] • Target: Incenter Angle Formula • Cross-Lesson: Topic 2.01 (Linear Angle Systems)

In ΔABC, angle bisectors of $\angle B$ and $\angle C$ intersect at the incenter I. If $\angle A = 68^\circ$, what is the measure of $\angle BIC$?

a. 112°

b. 124°

c. 134°

d. 146°

44 [Challenging] • [MCQ] • Target: Crossed-Ladders Similar Triangles Theorem • Cross-Lesson: Topic 10.01 (Harmonic & Rational Equations)

Two poles of heights $10\,\text{m}$ and $15\,\text{m}$ stand vertically on level ground $20\,\text{m}$ apart. A taut wire connects the top of each pole to the base of the opposite pole. At what height above the ground do the two wires intersect?

a. $5\,\text{m}$

b. $6\,\text{m}$

c. $6.25\,\text{m}$

d. $7.5\,\text{m}$

45 [Challenging] • [MCQ] • Target: Coordinate Triangle Area via Bounding Box / Shoelace • Cross-Lesson: Topic 1.01 (Coordinate Plane)

What is the area of the triangle with vertices $(0, 0)$, $(6, 2)$, and $(3, 8)$ in the xy-plane?

a. 18

b. 21

c. 24

d. 27

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